

Bayesian Statistics and Reinforcement Learning II

Belief states, Bayesian deep RL, and scientific applications

FYS4411 – Reinforcement Learning and Agentic AI

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Lecture 2 motivation

Lecture 1 introduced Bayesian inference inside RL: priors, posteriors, transition uncertainty, Thompson sampling, and UCB. Lecture 2 asks what happens when the posterior itself becomes part of the decision problem.

Takeaway. A Bayesian agent acts in the physical environment and in its own space of beliefs.

Learning goals

After this lecture you should be able to explain:

- ▶ the Bayes-adaptive viewpoint and belief states;
- ▶ why exact Bayesian RL is computationally hard;
- ▶ how posterior sampling and optimism approximate Bayesian planning;
- ▶ uncertainty-aware model-free and deep RL;
- ▶ applications in quantum engineering, MD, and robotics.

Recap: prior to posterior

Bayesian RL begins from a prior over unknown quantities:

$$P(M).$$

Experience is collected through interaction:

$$D_t = \{(S_k, A_k, R_{k+1}, S_{k+1})\}_{k=0}^{t-1}.$$

Bayes' rule gives

$$P(M \mid D_t) = \frac{P(D_t \mid M)P(M)}{P(D_t)}.$$

The posterior summarizes what the agent currently knows.

The missing state variable: belief

In an ordinary MDP, the current state S_t should contain all information needed to predict the future, given an action. In Bayesian RL, the model is unknown, so S_t alone is not enough. The agent also needs its current belief:

$$P(M \mid D_t).$$

A Bayesian information state is therefore

$$(S_t, P(M \mid D_t)).$$

Belief state interpretation

The pair

$$B_t = (S_t, P(M | D_t))$$

contains:

- ▶ where the agent is in the environment;
- ▶ what the agent currently believes about the environment;
- ▶ how confident it is in those beliefs.

Takeaway. The state of knowledge becomes part of the state for decision-making.

Actions change two things

When the agent chooses A_t :

1. the physical state changes:

$$S_t \rightarrow S_{t+1};$$

2. the belief changes:

$$P(M \mid D_t) \rightarrow P(M \mid D_{t+1}).$$

The new data point is

$$D_{t+1} = D_t \cup \{(S_t, A_t, R_{t+1}, S_{t+1})\}.$$

Takeaway. An action has both a physical effect and an informational effect.

The Bayes-adaptive viewpoint defines a larger decision process whose states are belief states:

$$B_t = (S_t, P(M \mid D_t)).$$

The transition in this larger process includes:

- ▶ sampling the next physical state and reward;
- ▶ applying Bayes' rule to update the posterior.

Takeaway. Bayesian RL treats learning itself as part of the sequential decision problem.

Optimal Bayesian action

An optimal Bayesian agent should choose actions that maximize expected return while accounting for how actions affect future information. Thus the value of an action includes:

- ▶ immediate reward;
- ▶ future rewards through the next physical state;
- ▶ information gained;
- ▶ improved future decisions due to a better posterior.

Why exact Bayesian RL is hard

Exact Bayes-adaptive planning requires reasoning over:

- ▶ future states;
- ▶ future rewards;
- ▶ future observations;
- ▶ future posterior distributions;
- ▶ policies that depend on future beliefs.

Even if the original MDP is finite, the belief space can be large or continuous.

Value of information

An action with low immediate reward can still be valuable if it gives information that improves future choices. Bayesian RL makes this idea explicit:

$$\text{action value} = \text{reward value} + \text{information value}.$$

The hard part is that information value depends on how future policies will use future beliefs

Discussion. When should a scientific experiment be performed even if it does not immediately optimize the objective?

Bayes-adaptive Bellman idea

Informally, a Bayes-adaptive value function is a value over belief states:

$$V(B_t) = V(S_t, P(M \mid D_t)).$$

A Bellman-style recursion would evaluate actions by expected reward plus discounted value of the next belief state:

$$V(B_t) = \max_a \mathbb{E} [R_{t+1} + \gamma V(B_{t+1}) \mid B_t, a].$$

The expectation includes uncertainty over models, transitions, rewards, and posterior updates.

Approximation map

Since exact Bayes-adaptive planning is difficult, practical methods approximate it:

Approximation	What it does
Posterior mean	collapse posterior to one model
Posterior sampling	sample one plausible model or value function
UCB/optimism	add bonus to uncertain actions
Ensembles	approximate posterior by multiple hypotheses
Probabilistic models	predict both mean and uncertainty

Posterior sampling as approximation

Posterior sampling approximates Bayesian planning by replacing the full belief-state problem with one sampled hypothesis:

$$M_t \sim P(M \mid D_t).$$

Then solve M_t as if it were true. This avoids explicit planning over future beliefs, but still uses posterior uncertainty to guide behavior.

Optimism as approximation

Optimism avoids sampling by acting as if uncertain quantities may take favorable values. For value-based action selection:

$$Q_{\text{opt}}(s, a) = \hat{Q}(s, a) + b(s, a).$$

For model-based planning, one may add an exploration bonus to rewards:

$$\hat{r}(s, a) + b(s, a).$$

The bonus should shrink as evidence accumulates.

Upper-confidence value iteration

An optimistic model-based update can be written schematically as

$$V_{k+1}(s) = \max_{a \in \mathcal{A}(s)} \left[\hat{r}(s, a) + b(s, a) + \gamma \sum_{s'} \hat{p}(s' | s, a) V_k(s') \right].$$

The bonus makes poorly understood actions temporarily attractive. If they are useful, the agent discovers value. If not, the bonus shrinks.

Posterior sampling versus optimism

Feature	Posterior sampling	Optimism/UCB
Uncertainty representation	posterior samples	confidence bonus
Action rule	solve sampled model	solve optimistic model
Exploration source	plausible worlds	upper bounds
Failure mode	bad posterior approximation	badly calibrated bonus

Model-free point estimates

Standard model-free RL often stores one number:

$$V(s) \quad \text{or} \quad Q(s, a).$$

This number estimates expected return, but it does not reveal how certain the agent is. Two state-action pairs may have the same Q estimate:

- ▶ one based on many observations;
- ▶ one based on one lucky reward.

Point estimates hide that distinction.

Posterior over action values

A Bayesian or uncertainty-aware model-free method conceptually represents

$$P(Q(s, a) \mid D_t).$$

This distribution describes plausible values of the state-action pair after observing data. If the posterior is broad, the agent should not act as if the value is known.

Takeaway. Bayesian model-free RL asks not only what the value is, but how well supported it is.

Uncertainty bonuses in model-free RL

A simple uncertainty-aware action rule is

$$A_t \in \arg \max_a [\hat{Q}(S_t, a) + b(S_t, a)] .$$

The bonus may depend on:

- ▶ visit counts;
- ▶ posterior variance;
- ▶ ensemble disagreement;
- ▶ prediction error or novelty.

This is not necessarily exact Bayesian inference, but it uses the same principle.

Ensembles as approximate posteriors

Train several value functions:

$$Q_1, Q_2, \dots, Q_K.$$

They may differ due to initialization, bootstrap data, randomized priors, or stochastic training. If the ensemble agrees, prediction is more reliable. If it disagrees, the agent has evidence of epistemic uncertainty.

Takeaway. An ensemble is a finite collection of plausible value functions.

Optimistic ensemble action rule

For neural action-value approximators $\hat{q}_i(s, a)$, use

$$\frac{1}{K} \sum_{i=1}^K \hat{q}_i(s, a) + \lambda \text{Std}(\hat{q}_1(s, a), \dots, \hat{q}_K(s, a)).$$

The mean estimates value. The standard deviation estimates uncertainty. The parameter λ controls exploration. This is a deep-RL analogue of upper-confidence action selection.

Posterior sampling with ensembles

A Thompson-like ensemble method:

1. at the beginning of an episode, sample an index $i \in \{1, \dots, K\}$;
2. act greedily with respect to Q_i for the episode;
3. update the ensemble from experience.

This samples a hypothesis instead of sampling an exact posterior.

Takeaway. Approximate posterior sampling can be implemented with bootstrapped value networks.

Uncertainty in Monte Carlo returns

Monte Carlo estimates a value by averaging returns. For a state-action pair,

$$\hat{Q}(s, a) = \frac{1}{N(s, a)} \sum_{i=1}^{N(s, a)} G^{(i)}(s, a).$$

A Bayesian view also asks about uncertainty in the mean return. With few returns, uncertainty about the mean is large. With many returns, it shrinks. This is the same conceptual move as Beta-Bernoulli or Dirichlet-categorical updating.

Deep RL and point-estimate networks

Deep RL often trains one network:

$$\hat{q}(s, a; w), \quad \hat{v}(s; w), \quad \pi_{\theta}(a \mid s).$$

After training, the network makes predictions, but it does not automatically tell us when those predictions are unsupported by data. This is dangerous because neural networks can be confidently wrong, especially off distribution.

Bayesian neural-network view

A Bayesian value network would maintain

$$P(w \mid D_t),$$

so predictions become distributions over functions. A Bayesian policy network could maintain

$$P(\theta \mid D_t).$$

Exact Bayesian inference over deep networks is usually intractable, so practical methods use approximations such as ensembles, dropout, variational inference, or probabilistic dynamics models.

Uncertainty-aware dynamics models

In model-based deep RL, a learned dynamics model may predict

$$(\hat{S}_{t+1}, \hat{R}_{t+1}) = f_w(S_t, A_t).$$

An uncertainty-aware version predicts not only a mean outcome, but also uncertainty:

- ▶ ensemble of dynamics models;
- ▶ probabilistic neural network;
- ▶ latent-variable dynamics model;
- ▶ posterior over model parameters.

Aleatoric and epistemic uncertainty in deep RL

Aleatoric

- ▶ irreducible randomness;
- ▶ e.g. sensor noise, stochastic dynamics;
- ▶ model should predict a distribution.

Epistemic

- ▶ reducible lack of knowledge;
- ▶ e.g. unfamiliar states;
- ▶ drives exploration or caution.

Takeaway. Bayesian RL is especially concerned with epistemic uncertainty.

Exploration or caution?

Uncertainty has two possible meanings for action selection:

- ▶ **exploratory mode:** try uncertain actions because they might be valuable;
- ▶ **safety mode:** avoid uncertain actions because consequences may be dangerous.

The same posterior can support either behavior depending on the task, cost, and safety constraints.

Discussion. In robotics, when should uncertainty encourage exploration and when should it trigger caution?

Bayesian DRL versus standard DRL

Standard deep RL	Bayesian/uncertainty-aware deep RL
one network prediction exploration by ϵ , entropy, noise confidence often implicit simple and scalable	distribution or ensemble of predictions exploration by uncertainty confidence explicitly represented more costly, more informative

Revisiting the Heisenberg analogy

A careful comparison:

- ▶ The Bayesian view is a different “picture” in the sense that it changes what we track.
- ▶ The standard picture tracks values, policies, and sampled updates.
- ▶ The Bayesian picture tracks beliefs over unknown quantities and updates those beliefs.
- ▶ But it is not a pure change of representation with identical algorithms. It can change behavior, computation, and approximations.

Takeaway. Think “belief-state picture,” not exact Heisenberg equivalence.

What changes across pictures?

Question	Standard RL	Bayesian RL
What is learned?	Q, V, π	posterior over M, Q, V, π, w
What is exploration?	added randomness	uncertainty resolution
What is state?	physical state S_t	belief state $(S_t, P(M D_t))$
What is difficult?	instability, sample efficiency	inference and planning over beliefs

Quantum engineering application I

Quantum devices often have uncertain Hamiltonian terms, noise rates, calibration drift, and partial measurement access. Bayesian RL formulation:

- ▶ model M : Hamiltonian, Lindblad parameters, readout errors;
- ▶ state/belief: measurements plus posterior over device parameters;
- ▶ action: pulse, detuning, measurement, feedback gate;
- ▶ reward: fidelity, energy, logical lifetime, sensing precision.

Quantum engineering application II

The Bayesian advantage is experimental economy. Quantum experiments can be expensive, noisy, and destructive. A Bayesian agent can choose experiments that are both useful for the control objective and informative about device uncertainty. Examples:

- ▶ pulse calibration with uncertain drift;
- ▶ feedback control under partial observations;
- ▶ adaptive measurements for sensing;
- ▶ active learning of noise models for error correction.

MD application I: active learning force fields

In molecular dynamics, learned force fields need to know when they are extrapolating. A Bayesian/uncertainty-aware workflow:

1. train a surrogate potential or force model;
2. run MD with uncertainty estimates;
3. identify uncertain configurations;
4. query expensive reference calculations;
5. update the model.

Takeaway. Uncertainty chooses where the next expensive calculation should be spent.

MD application II: enhanced sampling

Rare events and metastability make sampling difficult. Bayesian RL suggests an adaptive sampling policy:

- ▶ state: collective variables, uncertainty estimates, trajectory history;
- ▶ action: choose temperature, bias, umbrella window, launch state, or simulation length;
- ▶ reward: new states, uncertainty reduction, transition discovery, low bias cost;
- ▶ posterior: uncertainty over free energies, transition rates, or force-field predictions.

Robotics application I: dynamics uncertainty

Robots face model mismatch:

- ▶ unknown friction;
- ▶ contact discontinuities;
- ▶ actuator delay;
- ▶ changing payloads;
- ▶ sensor noise.

Bayesian RL can maintain a posterior or ensemble over dynamics models, then plan actions that balance performance, information, and safety.

Robotics application II: safe exploration

In robotics, exploration can damage hardware or surroundings. Uncertainty-aware control can support:

- ▶ optimistic exploration in simulation;
- ▶ conservative action choice on hardware;
- ▶ switching between learned policy and safe controller;
- ▶ uncertainty-triggered data collection;
- ▶ sim-to-real adaptation through posterior updating.

Takeaway. Bayesian information can decide when the agent should be bold and when it should be cautious.

Scientific experiments as Bayesian RL

Many scientific workflows are sequential decision problems:

- ▶ choose an experiment;
- ▶ observe data;
- ▶ update beliefs;
- ▶ choose the next experiment.

In this framing, the reward may be physical performance, information gain, cost reduction, or a combination. Bayesian RL naturally connects experimental design with control.

Partial observability and belief states

In many physical systems, the agent cannot observe the full Markov state. It observes

$$O_t \sim p(o \mid S_t).$$

A belief state may then include uncertainty over both:

- ▶ the current physical state;
- ▶ the model parameters.

This connects Bayesian RL to POMDPs, filtering, and state estimation.

Priors: helpful or dangerous?

Priors encode assumptions. They can improve data efficiency, but they can also bias learning. Good priors may come from:

- ▶ symmetry;
- ▶ conservation laws;
- ▶ known physics;
- ▶ simulation data;
- ▶ previous experiments.

Bad priors can slow learning or hide surprising behavior.

Takeaway. In physics, priors are often where domain knowledge enters.

Reward versus information

A Bayesian agent may care about both task reward and information. A shaped objective might include

$$r_t^{\text{total}} = r_t^{\text{task}} + \eta r_t^{\text{info}} - c_t.$$

Examples of information rewards:

- ▶ posterior entropy reduction;
- ▶ reduction in model variance;
- ▶ disagreement reduction;
- ▶ expected improvement in future control.

But adding information rewards can change the task, so use them carefully.

Computational costs

Bayesian RL can be expensive because it may require:

- ▶ maintaining posteriors;
- ▶ sampling models or parameters;
- ▶ planning in many sampled models;
- ▶ training ensembles;
- ▶ estimating uncertainty reliably;
- ▶ recalibrating beliefs after distribution shift.

Takeaway. The cost is the price of knowing what the agent does not know.

Diagnostics for uncertainty-aware RL

Useful diagnostics include:

- ▶ regret curves for bandits;
- ▶ calibration of uncertainty estimates;
- ▶ posterior or ensemble spread over time;
- ▶ failure rate under out-of-distribution states;
- ▶ performance under equal sample budget;
- ▶ behavior when uncertainty is high;
- ▶ sensitivity to prior choice and reward weights.

Common pitfalls

- ▶ treating aleatoric noise as reducible epistemic uncertainty;
- ▶ using a posterior approximation that is poorly calibrated;
- ▶ adding uncertainty bonuses so large that the task reward is ignored;
- ▶ using priors that encode the answer too strongly;
- ▶ comparing Bayesian and non-Bayesian agents with unequal compute budgets;
- ▶ equating uncertainty-aware heuristics with exact Bayesian inference.

Mini case: bandit comparison

A useful classroom experiment:

- ▶ five Bernoulli arms with probabilities $[0.15, 0.30, 0.45, 0.55, 0.60]$;
- ▶ compare ϵ -greedy, UCB1, Bayesian UCB, and Thompson sampling;
- ▶ plot cumulative pseudo-regret;
- ▶ run many seeds;
- ▶ ask: which method explores too much, too little, or just enough?

Mini case: Bayesian gridworld

A small MDP version:

- ▶ gridworld with unknown slip probability;
- ▶ prior over slip probability;
- ▶ actions reveal transitions;
- ▶ posterior over dynamics is updated from counts;
- ▶ compare posterior-mean planning and posterior sampling.

Discussion. How does behavior change when the agent is uncertain about whether the grid is slippery?

Mini project: uncertainty-aware DQN

Extend a simple DQN example:

- ▶ train an ensemble of K Q-networks;
- ▶ use mean plus standard deviation for action selection;
- ▶ compare with ϵ -greedy exploration;
- ▶ measure performance, failures, and exploration patterns;
- ▶ test on states outside the training distribution.

This gives a practical bridge from Bayesian ideas to deep RL.

Discussion questions

1. What is the posterior over in a quantum control experiment?
2. When does information have reward value?
3. Is Thompson sampling more Bayesian than UCB?
4. Can a deterministic policy still be Bayesian?
5. What should a robot do when its uncertainty is high?
6. How would you detect that an ensemble is overconfident?

Lecture 2 summary

- ▶ The Bayes-adaptive viewpoint treats belief as part of the decision state.
- ▶ Actions change both the physical state and the posterior belief.
- ▶ Exact Bayesian RL is hard because it requires planning over future beliefs.
- ▶ Posterior sampling, optimism, ensembles, and probabilistic models are practical approximations.
- ▶ Applications in quantum engineering, MD, and robotics are driven by the same need: act under limited data and uncertainty.

References and reading

- ▶ Course notes, Chapter 4: Bayesian Reinforcement Learning and Uncertainty.
- ▶ Sutton and Barto: bandits, exploration, and model-based RL.
- ▶ Xiao: Thompson sampling and UCB implementation examples.
- ▶ Lockwood and Si, *A Review of Uncertainty for Deep Reinforcement Learning*, AIIDE 2022.
- ▶ Xie et al., Bayesian active learning for uncertainty-aware molecular dynamics, 2022.
- ▶ Rana et al., Bayesian controller fusion for robotics, IJRR 2023.